A Universal Phase-Sector Selector for Fixed-Shift Coherent Prime Fibers: Constant-Energy Extraction, Adaptive Necessity, and the Exact Physical Gate

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Abstract

For the literal fixed-shift packet, the terminal map first restricts the full pre-terminal array to the exact prime equality support, multiplies a surviving atom by its cofactor sign and shaped terminal logarithm, and then sums distinct terminal primes inside a complete output fiber $y = (\gamma, d(m)r, j)$. TPC–55 proved that adjoining the terminal prime to the output key removes every collision on one declared source system. The remaining internal gate is therefore the coherent square of one explicit prime-indexed vector in each fiber. That square can vanish even when its terminal diagonal is positive.

We prove a universal lower-selection theorem for this exact gate. Fix a unit gauge on each output fiber and partition the gauge-normalized realized nonzero amplitudes into $M \geq 3$ equal angular sectors. One global sector index in these normalized coordinates, used simultaneously on every fiber, selects a coordinate subpacket whose grouped energy is at least

$$\frac{\cos^2(\pi/M)}{M}$$

times the full terminal diagonal energy. In each fiber the selected atoms lie, after its own fixed gauge, in the same relative cone of half-angle $\pi/M < \pi/2$. The corresponding absolute cones may rotate with the fiber, and the estimate is independent of the fiber cardinalities. Among equal finite sectorizations, $M = 5$ is optimal and gives the exact constant $(3 + \sqrt{5})/40$. Allowing a continuously rotated angular window gives the slightly larger constant $c_* = \theta_* \cos^2 \theta_*/\pi = 0.1311275\ldots$, where $2\theta_* \tan \theta_* = 1$. We also prove a quantitative phase-error version.

For the literal physical vector, the common gauge is the exact product of the residual Möbius sign, opposite-row phase, orbit factor, and common opposite-row profiles. The terminal-prime bijection makes every selected sector an unambiguous coordinate projection on the actual terminal support. Consequently, conditional on a separately established terminal-activation ratio, the adaptive pentant subpacket has no positive polynomial coherence loss at this selected grouping gate. Relative to the earlier rank-by-prime selection, it improves the retained-energy constant at that gate, at the price of coefficient adaptivity: it retains a constant fraction of the terminal diagonal without a maximal-multiplicity factor or a balanced cofactor hypothesis.

The qualifier *adaptive* is essential for the universal all-coordinate guarantee. We prove that any selector fixed independently of the coefficient phases and retaining two coordinates in one fiber has a nonzero collision-kernel direction on the unrestricted coordinate space. The inherited class with only an upper bound on the complex source factor can realize that cancellation locally on any nondegenerate two-atom literal carrier; this is not promoted to a global literal-class no-go. Thus the theorem is a fixed-h_0, post-terminal selected-square

result, not a lower bound for the undeleted output. It does not prove terminal activation, a coefficient-independent pre-terminal phase realization, frozen-shift localization, native-Gram comparison, a parity improvement, or a twin-prime statement.

Keywords: coherent terminal-prime square; phase-sector selection; residual fiber; fixed physical shift; collision kernel; parity boundary.

Convention. One nonzero physical shift h_0, one left source system, and one direct residue/orbit slice are fixed. All domain and output spaces use the raw counting-measure atomic normalization inherited from TPC–53. The shaped terminal weight $\Lambda(n) = (\log n)\Phi_0(n/X)$ is displayed in the operator and is not the von Mangoldt function. A selector constructed from the realized terminal phases is called *adaptive*; this word is never suppressed when the selector is used.

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1 Introduction

The preceding three stages of this program separated three losses that had previously been hidden inside one schematic prime square. TPC–53 fixed the literal four-factor coefficient dictionary and the raw atomic normalization [4]. TPC–54 proved sharp pre-terminal energy selection laws and showed that magnitude information cannot survive an arbitrary many-to-one complex sum [5]. TPC–55 then resolved the complete fixed-shift output fiber [6]. If

$$u = (m, r, j, \gamma), \qquad m = d\ell,$$

survives the exact terminal restriction, then

$$p_u = \frac{mj + h_0}{r} \tag{1}$$

is prime and the complete output is

$$y = q(u) = (\gamma, s = dr, j). \tag{2}$$

Under the inherited row geometry, adjoining p_u to the key is injective. Thus the only remaining collision is the deliberate operation that forgets p and coherently adds all terminal primes in the same (γ, s, j)-fiber.

Write the terminalized amplitude indexed by that prime as

$$x_{y,p} = \mu(r_{y,p})\Lambda(pr_{y,p})g_{u(y,p)}. \tag{3}$$

The exact diagonal and grouped energies are

$$\mathcal{D}_{h_0}(g) = \sum_y \sum_{p\in\mathcal{P}_y} |x_{y,p}|^2, \qquad \mathcal{G}_{h_0}(g) = \sum_y \left| \sum_{p\in\mathcal{P}_y} x_{y,p} \right|^2. \tag{4}$$

All sums in (4) are over the exact finite terminal support; $\mathcal{P}_y$ is not an ambient interval of primes. The ratio $\mathcal{G}_{h_0}/\mathcal{D}_{h_0}$ can be zero. TPC–55 supplied two ways to avoid that zero: impose an acute sector condition, or retain at most one prime in each fiber. The first was only a criterion. The second was an unconditional post-terminal construction, but on a cofactor block it retained only the reciprocal of the maximal fiber multiplicity.

The present paper asks a narrower question:

> Can one always extract a genuinely coherent terminal-prime subpacket while retaining a constant fraction of the full terminal diagonal, independently of the fiber sizes?

The answer is yes if the selection is permitted to use the realized phases. The proof is elementary, but its placement in the physical operator is order-sensitive. Split the complex plane into five equal sectors after removing the common output gauge. The five sectors partition the terminal diagonal, so one of them carries at least one fifth of that energy. Inside one sector every amplitude has real projection at least $\cos(\pi/5)$ times its modulus. Therefore the selected grouped square is at least

$$c_5\mathcal{D}_{h_0}(g), \qquad c_5 := \frac{\cos^2(\pi/5)}{5} = \frac{3+\sqrt{5}}{40}. \tag{5}$$

Only one sector index is selected globally in the gauge-normalized phase coordinates; it is not chosen separately for every output fiber. The corresponding absolute angular window is rotated by the canonical gauge of that fiber.

This gives a real advance at the selected grouping gate. No X^{ε}, divisor, or maximal-multiplicity factor appears. If a pre-terminal packet satisfies the separate activation estimate

$$\mathcal{D}_{h_0}(g) \ge X^{-\lambda_{\mathrm{term}}+o(1)}\mathcal{E}_{\mathrm{pre}}(g), \tag{6}$$

then the selected physical output satisfies

$$\mathcal{G}_{h_0}(P_*g) \ge c_5 X^{-\lambda_{\mathrm{term}}+o(1)}\mathcal{E}_{\mathrm{pre}}(g). \tag{7}$$

Thus the adaptive coherence exponent is zero. In contrast with the rank-by-prime transversal, no balanced condition such as $R \gg L$ is needed.

There is an equally important negative half. The sets selected by the five-sector rule depend on the phases of $x_{y,p}$. Once the realized vector is fixed, each is an ordinary orthogonal coordinate projection; as a rule on a coefficient class, however, the construction is nonlinear. On the unrestricted coordinate space, if a coefficient-independent set contains two coordinates of one fiber, the vector assigning opposite values to those coordinates lies in the grouping kernel. The inherited hypothesis $|\zeta_m| \ll 1$ also realizes this cancellation *inside* any nondegenerate two-atom literal fiber by sufficiently small choices of the two relevant source factors. This proves failure of a uniform fiberwise lower bound for such a selector. It does not by itself disprove a global lower bound over the coupled literal coefficient class, because the same row coefficients may activate other fibers. A global literal-class no-go would require control or isolation of those additional outputs.

This dichotomy is useful for the route ahead:

$$\boxed{\begin{array}{c}\text{adaptive terminal phases} \Rightarrow \text{constant-energy coherent subpacket,}\\ \text{uniform selection on unrestricted arrays} \Rightarrow \text{at most one atom per fiber,}\end{array}} \tag{8}$$

It turns an unstructured prime-collision space into a five-channel phase interface. A future paper can now ask whether the five channels come from fixed residue and sign-stable source cells, or whether every such attempted pre-terminal realization meets a collision obstruction, with global literal-class saturation requiring separate control of row couplings.

1.1 Relation to the earlier coherent square

The physical object in (4) must not be identified with the projective/frozen square isolated in TPC–48 [3]. That earlier object has the schematic form

$$\sum_{m,r}\left\|\sum_{p} A_\omega(p,r,m)e(-pr\alpha)v_{\omega,p,r,m}\right\|^2, \tag{9}$$

before signed projective reassembly. It keeps (m,r) separate and extends the physical coefficient field to frozen intercepts. The present square is already reassembled, uses one specified h_0, and groups by the complete literal output $(\gamma, d(m)r, j)$. A lower estimate for an average of (9) does not become a lower estimate for (4) without separate lower reassembly and fixed-shift localization theorems.

1.2 Organization and claim level

Section 2 states the full physical coefficient and support dictionary. Section 3 proves the abstract finite-sector theorem, its optimal equal-sector constant, the rotating window refinement, and phase stability. Section 4 specializes it through the exact terminal-prime bijection. Section 5 proves adaptive necessity and gives a pre-terminal phase-realization criterion. Section 6 records the frozen-average and signed-reassembly non-implications. Section 7 gives the endpoint ledger and six-point audit.

The sector theorem is unconditional finite-dimensional mathematics (L0). Its use on the literal fixed-h_0 terminal vector is an exact physical interface result (L1). Terminal activation and a pre-terminal realization of the adaptive selector are not proved. Nothing here improves a sieve lower bound or crosses the parity boundary described in the classical sieve literature [1]; no prime-pair or twin-prime conclusion is asserted.

2 The literal fixed-shift terminal-prime interface

This section repeats the complete coefficient and support information needed for the sector theorem. The purpose is to prevent a lower bound for a renormalized model, an unrestricted prime ladder, or a frozen intercept from being substituted for the physical map.

2.1 Rows, masks, and the full pre-terminal universe

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$, put $H = \operatorname{rad}|h_0|$, and use one left source system. A direct row and an opposite row are

$$m = d_m \ell_m, \qquad m_\gamma = d_\gamma \ell_\gamma, \tag{10}$$

where the ℓ's are prime, the d's are squarefree,

$$\ell_m, \ell_\gamma \in [L, 2L], \qquad d_m, d_\gamma \in [D, 2D], \qquad (d_m d_\gamma, H) = 1. \tag{11}$$

The inherited separation makes each integer row determine its displayed factorization. The scales are

$$Q = LD, \qquad K = DJ, \qquad QJ \asymp X, \qquad \kappa_0 = \frac{1}{400}, \qquad \Delta = QX^{-\kappa_0}, \qquad G = X^{\kappa_0}. \tag{12}$$

Fix one direct residue group, one orbit class $a \pmod{H_{\mathrm{orb}}}$ with $H_{\mathrm{orb}} = O_{h_0}(1)$, one compact opposite-row box, and one compact orbit interval. None of these fixed choices has positive polynomial cost.

The exact source coefficient is

$$a_m = \mu(d_m)(\log \ell_m)\omega_D(d_m)\psi_L(\ell_m/L)\zeta_m, \qquad |\zeta_m| \ll 1. \tag{13}$$

Here $\zeta_m = \zeta_m^{(1)}$ is the inherited bounded complex source factor; no lower bound or phase condition is assumed. The opposite row contributes

$$\eta_\gamma \mathscr{R}(q_\gamma), \qquad q_\gamma = \left(\log \frac{\ell_\gamma}{L}, \log \frac{d_\gamma}{D}\right), \qquad |\eta_\gamma| = 1, \tag{14}$$

and the fixed orbit cell contributes $\xi_a \neq 0$. The literal mask and the declared four-factor profile are

$$K_{m\gamma} = \mathbf{1}_{\ell_m \neq \ell_\gamma} \mathbf{1}_{|m - m_\gamma| > \Delta} \mathbf{1}_{(d_m, d_\gamma) \leq G}, \tag{15}$$

$$\mathcal{W}_{m,\gamma}(j/J) = W_1(mj/X)\Psi_1(d_m j/K)W_2(m_\gamma j/X)\Psi_2(d_\gamma j/K). \tag{16}$$

Every smooth function in these formulas belongs to the fixed bounded families declared in TPC–53 [4].

Let $\mathcal{I}_X$ be the exact retained set of (m,r)-pairs, $\mathcal{O}_X$ the complete opposite-row cell, and

$$\mathcal{J}_{X,a} = \{j : j/J \in I_0,\ j \equiv a \pmod{H_{\mathrm{orb}}}\}. \tag{17}$$

The full pre-terminal atomic universe and physical pre-terminal vector are

$$\mathcal{U}^{\mathrm{pre}} = \{(m,r,j,\gamma) : (m,r) \in \mathcal{I}_X, j \in \mathcal{J}_{X,a},\ \gamma \in \mathcal{O}_X\}, \tag{18}$$

$$[G_X^L]_u = a_m K_{m\gamma}\eta_\gamma \mathscr{R}(q_\gamma)\xi_a \mathcal{W}_{m,\gamma}(j/J), \qquad u = (m,r,j,\gamma). \tag{19}$$

The repeated pre-terminal value in (19) has no r-dependence. Its repetitions are distinct counting-measure atoms.

2.2 Terminal restriction and the prime-resolved key

Let

$$\Lambda(n) = (\log n)\Phi_0(n/X). \tag{20}$$

This is a shaped terminal logarithm, not the von Mangoldt function. For $u = (m,r,j,\gamma) \in \mathcal{U}^{\mathrm{pre}}$, set

$$p_u = \frac{mj + h_0}{r} \tag{21}$$

when the quotient is integral. The exact terminal-admissible universe is

$$\begin{aligned}\mathcal{U}_{h_0}^{\mathrm{term}} := \{u \in \mathcal{U}^{\mathrm{pre}} : {}& p_u \text{ is prime in the inherited terminal support},\\ & p_u r - mj = h_0, \quad \Lambda(p_u r) \neq 0,\\ & d_m, r, d_m r \text{ are squarefree}, \quad (d_m, r) = 1\}.\end{aligned} \tag{22}$$

The actually nonzero physical support is the separate set

$$\mathcal{U}_{h_0}^{\mathrm{phys}} = \{u \in \mathcal{U}_{h_0}^{\mathrm{term}} : [G_X^L]_u \neq 0\}. \tag{23}$$

On $\mathcal{U}_{h_0}^{\mathrm{term}}$, the inherited equality support gives

$$s = d_m r \text{ squarefree}, \qquad (d_m, r) = 1, \qquad \mu(d_m)\mu(r) = \mu(s). \tag{24}$$

The grouped and refined keys are

$$q(u) = (\gamma, d_m r, j), \qquad \widetilde{q}(u) = (\gamma, d_m r, j, p_u). \tag{25}$$

TPC–55 proves the following exact interface under the inherited geometry [6].

Proposition 2.1 (Inherited prime-resolved bijection)**.** *Assume $h_0 \neq 0$, $D = o(J)$, uniqueness of the row factorization, and sufficiently large X. Then $\widetilde{q}$ is injective on $\mathcal{U}_{h_0}^{\mathrm{term}}$ for one declared source system. For every $y = (\gamma, s, j)$, the map $u \mapsto p_u$ is therefore a bijection*

$$\mathcal{F}_y^{\mathrm{term}} := q^{-1}(y) \cap \mathcal{U}_{h_0}^{\mathrm{term}} \longleftrightarrow \mathcal{P}_y^{\mathrm{term}} := \{p_u : u \in \mathcal{F}_y^{\mathrm{term}}\}, \tag{26}$$

and restricts to a bijection between the corresponding physical subsets.

Explicitly, the latter subsets are

$$\mathcal{F}_y^{\mathrm{phys}} := q^{-1}(y) \cap \mathcal{U}_{h_0}^{\mathrm{phys}}, \qquad \mathcal{P}_y^{\mathrm{phys}} := \{p_u : u \in \mathcal{F}_y^{\mathrm{phys}}\}, \tag{27}$$

and $u \mapsto p_u$ is a bijection between them. Thus $\mathcal{P}_y^{\mathrm{term}}$ is the exact finite set determined by the pre-terminal carrier together with the terminal equality, primality, terminal-weight, squarefreeness, and coprimality conditions. It does not impose nonvanishing of every literal mask, source, or profile factor. The set $\mathcal{P}_y^{\mathrm{phys}}$ imposes the additional condition $[G_X^L]_u \neq 0$, and therefore records the actually nonzero literal support. Replacing either set by all prime points on an ambient affine ladder would enlarge the corresponding operator.

The exact terminal and physical output index sets are

$$\mathcal{Y}_{h_0}^{\mathrm{term}} := q(\mathcal{U}_{h_0}^{\mathrm{term}}), \qquad \mathcal{Y}_{h_0}^{\mathrm{phys}} := q(\mathcal{U}_{h_0}^{\mathrm{phys}}). \tag{28}$$

2.3 Three energies and the common physical gauge

For an arbitrary array $g=(g_u)_{u\in\mathcal{U}^{\mathrm{pre}}}$, define the terminal diagonal on the full pre-terminal space by

$$(D_{h_0}g)_u=\begin{cases}\mu(r_u)\Lambda(p_ur_u)g_u, & u\in\mathcal{U}^{\mathrm{term}}_{h_0},\\ 0, & u\notin\mathcal{U}^{\mathrm{term}}_{h_0}.\end{cases}\tag{29}$$

The literal grouped map is

$$(\mathcal{T}_{h_0}g)_y=\sum_{u\in\mathcal{F}^{\mathrm{term}}_y}(D_{h_0}g)_u.\tag{30}$$

With counting measure throughout, put

$$\mathcal{E}_{\mathrm{pre}}(g)=\sum_{u\in\mathcal{U}^{\mathrm{pre}}}|g_u|^2,\tag{31}$$

$$\mathcal{D}_{h_0}(g)=\|D_{h_0}g\|_2^2=\sum_{u\in\mathcal{U}^{\mathrm{term}}_{h_0}}|\Lambda(p_ur_u)|^2|g_u|^2,\tag{32}$$

$$\mathcal{G}_{h_0}(g)=\|\mathcal{T}_{h_0}g\|_2^2=\sum_y\left|\sum_{u\in\mathcal{F}^{\mathrm{term}}_y}(D_{h_0}g)_u\right|^2.\tag{33}$$

The simplification in (32) uses squarefreeness of r on the terminal support. The terminal weight has not been hidden in the input norm.

For the distinguished physical vector, every nonzero terminal amplitude in a fiber has an exact common factor. If $u=(d\ell,r,j,\gamma)$ and $y=(\gamma,s,j)$, define

$$\Omega_y:=\mu(s)\eta_\gamma\mathscr{R}(q_\gamma)\xi_aW_2(m_\gamma j/X)\Psi_2(d_\gamma j/K),\tag{34}$$

$$b_u:=(\log\ell)\omega_D(d)\psi_L(\ell/L)\zeta_mK_{m\gamma}W_1(mj/X)\Psi_1(dj/K)\Lambda(mj+h_0).\tag{35}$$

Then (24) gives

$$(D_{h_0}G^L_X)_u=\Omega_yb_u.\tag{36}$$

If a fiber has positive terminal diagonal energy, then $\Omega_y\neq 0$. Its canonical unit gauge is therefore

$$\omega_y=\frac{\Omega_y}{|\Omega_y|}.\tag{37}$$

After multiplication by $\overline{\omega_y}$, the phases tested below are exactly the phases of the remaining literal amplitudes b_u. The common Möbius sign has disappeared, but no phase condition on b_u has been assumed.

3 A universal phase-sector lower selection theorem

We first remove all arithmetic. Let $\mathcal{U}$ be a finite set partitioned into fibers

$$\mathcal{U}=\bigsqcup_{y\in\mathcal{Y}}\mathcal{F}_y,\tag{38}$$

and let $x=(x_u)_{u\in\mathcal{U}}\in\mathbb{C}^{\mathcal{U}}$. The unweighted grouping map is

$$(\Sigma x)_y=\sum_{u\in\mathcal{F}_y}x_u.\tag{39}$$

Choose an arbitrary unit gauge $\omega_y\in\mathbb{C}$, $|\omega_y|=1$, on every fiber. The gauge family is part of the data and is fixed before the maximizing sector index is selected. These gauges only rotate output coordinates and hence do not change any grouped norm.

3.1 Equal sectors and one global choice

Fix an integer $M \geq 3$. On the phase circle use the half-open sectors

$$I_k = \left[\frac{2\pi k}{M} - \frac{\pi}{M}, \frac{2\pi k}{M} + \frac{\pi}{M}\right) \pmod{2\pi}, \qquad 0 \leq k < M. \tag{40}$$

For $x_u \neq 0$, put

$$\varphi_u = \operatorname{Arg}(\overline{\omega_{q(u)}} x_u) \pmod{2\pi}, \tag{41}$$

where $q(u) = y$ for $u \in \mathcal{F}_y$, and define

$$\mathcal{S}_k(x; \omega) = \{u : x_u \neq 0,\ \varphi_u \in I_k\}. \tag{42}$$

Let $P_k = P_k(x; \omega)$ be the orthogonal coordinate projection onto $\mathcal{S}_k(x; \omega)$. Zero coordinates can be assigned to any sector; they affect none of the identities. The notation records an important fact: after x is fixed, P_k is linear and orthogonal, but the rule $x \mapsto P_k(x; \omega)$ is nonlinear.

Define the selected diagonal and grouped energies

$$D_k(x) := \|P_k x\|_2^2, \qquad C_k(x) := \|\Sigma P_k x\|_2^2. \tag{43}$$

Theorem 3.1 (Equal-sector compression and selection)**.** *For every finite fiber system, every x, every fixed family of unit gauges, and every integer $M \geq 3$,*

$$\sum_{k=0}^{M-1} D_k(x) = \|x\|_2^2, \tag{44}$$

$$C_k(x) \geq \cos^2(\pi/M) D_k(x) \qquad (0 \leq k < M), \tag{45}$$

$$\sum_{k=0}^{M-1} C_k(x) \geq \cos^2(\pi/M)\|x\|_2^2. \tag{46}$$

Let k_* be the least index maximizing $D_k(x)$. Then this canonical tie rule gives

$$\boxed{\|\Sigma P_{k_*} x\|_2^2 \geq c_M \|x\|_2^2, \qquad c_M := \frac{\cos^2(\pi/M)}{M}.} \tag{47}$$

The same index k_* is used on every fiber in the normalized coordinates $\operatorname{Arg}(\overline{\omega_y} x_u)$. Since ω_y may vary with y, the corresponding absolute angular window is a fiber-dependent rotation.

Proof. The sector supports partition the nonzero coordinates, proving (44). Fix k and a fiber y. For every $u \in \mathcal{F}_y \cap \mathcal{S}_k$, one can write

$$\overline{\omega_y} e^{-2\pi i k/M} x_u = |x_u| e^{i\delta_u}, \qquad |\delta_u| \leq \frac{\pi}{M}, \tag{48}$$

with one endpoint understood by the half-open convention. Therefore

$$\begin{aligned} \left|\sum_{u \in \mathcal{F}_y \cap \mathcal{S}_k} x_u\right| &\geq \operatorname{Re}\left(\overline{\omega_y} e^{-2\pi i k/M} \sum_{u \in \mathcal{F}_y \cap \mathcal{S}_k} x_u\right) \\ &\geq \cos(\pi/M) \sum_{u \in \mathcal{F}_y \cap \mathcal{S}_k} |x_u| \\ &\geq \cos(\pi/M) \left(\sum_{u \in \mathcal{F}_y \cap \mathcal{S}_k} |x_u|^2\right)^{1/2}. \end{aligned} \tag{49}$$

Squaring and summing over y proves (45); summing over k gives (46). Finally, $D_{k_*} \geq M^{-1}\|x\|_2^2$, so (45) gives (47). □

The theorem contains two different objects. The phase-split synthesis

$$x \longmapsto \left(\sum_{u \in \mathcal{F}_y \cap \mathcal{S}_k} x_u \right)_{(y,k)} \tag{50}$$

has a lower bound on the distinguished vector used to define its sectors. The original output instead sums the M sector outputs at fixed y. Those final M channels may cancel. Thus (46) is not a lower bound for $\|\Sigma x\|_2$.

3.2 Why five equal sectors are best in this family

Proposition 3.2 (Optimal equal-sector count)**.** *Among integers $M \geq 3$, the certified constant $c_M = M^{-1}\cos^2(\pi/M)$ is uniquely maximized at $M = 5$. Its value is*

$$\boxed{c_5 = \frac{\cos^2(\pi/5)}{5} = \frac{3+\sqrt{5}}{40} = 0.1309016994\ldots.} \tag{51}$$

Proof. For real $t \geq 3$, let $f(t) = t^{-1}\cos^2(\pi/t)$. Direct differentiation gives

$$\frac{f'(t)}{f(t)} = \frac{2\pi\tan(\pi/t) - t}{t^2}. \tag{52}$$

After setting $\theta = \pi/t$, the unique critical point is determined by $2\theta\tan\theta = 1$. The left side is strictly increasing on $(0, \pi/2)$, and the corresponding t lies strictly between 4 and 5. Hence the integer maximum is either 4 or 5. Now

$$c_4 = \frac{1}{8}, \qquad c_5 = \frac{3+\sqrt{5}}{40} > \frac{1}{8}. \tag{53}$$

The monotonicity on either side of the critical point proves uniqueness. □

The word *optimal* in proposition 3.2 refers only to the constants certified by equal finite sectorizations. It is not a global optimality theorem over all nonlinear subset rules.

3.3 A continuously rotated window

The finite pentant library is convenient for a literal audit. If a continuum of rotations is allowed, the constant can be improved slightly. For $0 < \theta < \pi/2$ and $\alpha \in \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, retain the coordinates satisfying

$$\mathrm{dist}_{\mathbb{T}}(\varphi_u, \alpha) \leq \theta \tag{54}$$

and denote the associated coordinate projection by $P_{\alpha,\theta}$.

Theorem 3.3 (Rotating-window selection)**.** *For every $0 < \theta < \pi/2$, there exists $\alpha \in \mathbb{T}$ such that*

$$\|\Sigma P_{\alpha,\theta} x\|_2^2 \geq c(\theta)\|x\|_2^2, \qquad c(\theta) := \frac{\theta}{\pi}\cos^2\theta. \tag{55}$$

The function $c(\theta)$ has a unique maximizer $\theta_* \in (0, \pi/2)$ characterized by

$$2\theta_* \tan\theta_* = 1. \tag{56}$$

Numerically,

$$\theta_* = 0.6532711871\ldots, \qquad c_* = c(\theta_*) = 0.1311275284\ldots. \tag{57}$$

Proof. For each nonzero coordinate, the set of centers α satisfying (54) has normalized Haar measure θ/π. Fubini therefore gives

$$\int_{\mathbb{T}} \|P_{\alpha,\theta} x\|_2^2 \frac{d\alpha}{2\pi} = \frac{\theta}{\pi}\|x\|_2^2. \tag{58}$$

Some center retains at least that much diagonal energy. The proof of (49), with θ in place of π/M, proves (55). Finally,

$$c'(\theta) = \frac{\cos\theta}{\pi}(\cos\theta - 2\theta\sin\theta), \tag{59}$$

so the unique interior critical point satisfies (56); endpoint values are zero. $\square$

3.4 Stability under phase uncertainty

The sector theorem is not destroyed by small errors in the phases used to form the coordinate sets.

Theorem 3.4 (Phase-error stability)**.** *Suppose approximate normalized phases $\widehat{\varphi}_u$ satisfy*

$$\operatorname{dist}_{\mathbb{T}}(\widehat{\varphi}_u, \varphi_u) \le \varepsilon \qquad (x_u \neq 0). \tag{60}$$

Partition the coordinates by the M equal sectors of the approximate phases, and choose the least sector with maximal true diagonal energy. If

$$0 \le \varepsilon < \frac{\pi}{2} - \frac{\pi}{M}, \tag{61}$$

then the resulting projection $\widehat{P}_*$ satisfies

$$\boxed{\|\Sigma \widehat{P}_* x\|_2^2 \ge \frac{\cos^2(\pi/M + \varepsilon)}{M}\|x\|_2^2.} \tag{62}$$

Proof. The approximate sectors still partition the active coordinates, so the selected sector carries at least $M^{-1}\|x\|_2^2$ diagonal energy. Every true phase in that sector lies within $\pi/M + \varepsilon < \pi/2$ of its center. Apply the real-projection argument in (49) with this enlarged half-angle. $\square$

The error ε may include both an atomwise phase error and a fiberwise error in the chosen common gauge. The theorem requires exact magnitudes only to identify an energy-maximizing sector; a separate magnitude-approximation error would have to be charged explicitly.

4 Specialization to the literal terminal-prime square

We now apply theorem 3.1 to $x = D_{h_0} G_X^L$. The specialization uses the exact physical support twice: first to identify each coordinate with one terminal prime, and then to remove the common fiber gauge without changing any phase difference.

4.1 Prime-indexed pentants

For an active output y, let $u(y,p)$ be the unique input associated with $p \in \mathcal{P}_y^{\mathrm{phys}}$ by proposition 2.1, and put

$$x_{y,p} := (D_{h_0} G_X^L)_{u(y,p)} = \mu(r_{y,p})\Lambda(pr_{y,p})[G_X^L]_{u(y,p)}. \tag{63}$$

Then

$$D_y := \sum_{p \in \mathcal{P}_y^{\mathrm{phys}}} |x_{y,p}|^2, \tag{64}$$

$$C_y := \left| \sum_{p \in \mathcal{P}_y^{\mathrm{phys}}} x_{y,p} \right|^2. \tag{65}$$

Empty fibers have $D_y = C_y = 0$. A nonzero singleton fiber has $C_y = D_y$. Globally,

$$\mathcal{D}_{h_0}(G_X^L) = \sum_y D_y, \qquad \mathcal{G}_{h_0}(G_X^L) = \sum_y C_y. \tag{66}$$

For $D_y > 0$, use the literal unit gauge ω_y from (37). Equations (36) and (63) show that

$$\operatorname{Arg}(\overline{\omega_y} x_{y,p}) = \operatorname{Arg} b_{u(y,p)}. \tag{67}$$

Partition these exact phases into the five fixed intervals $I_0, \ldots, I_4$ of (40). Let

$$\mathcal{P}_{y,k}^{\mathrm{sec}} := \{p \in \mathcal{P}_y^{\mathrm{phys}} : \operatorname{Arg} b_{u(y,p)} \in I_k\}, \tag{68}$$

and pull this set back through the prime-resolved bijection. This defines an orthogonal coordinate projection P_k^{sec} on the terminal coordinate space. Extend it by zero to the rest of $\mathcal{U}^{\mathrm{pre}}$. For the fixed realized vector, diagonal coordinate projections commute with the terminal diagonal:

$$D_{h_0} P_k^{\mathrm{sec}} G_X^L = P_k^{\mathrm{sec}} D_{h_0} G_X^L. \tag{69}$$

Theorem 4.1 (Literal pentant extraction)**.** *Assume the hypotheses of proposition 2.1. Let k_* be the least pentant index whose selected terminal diagonal energy is maximal. Then*

$$\mathcal{D}_{h_0}(P_{k_*}^{\mathrm{sec}} G_X^L) \geq \frac{1}{5} \mathcal{D}_{h_0}(G_X^L), \tag{70}$$

$$\boxed{\mathcal{G}_{h_0}(P_{k_*}^{\mathrm{sec}} G_X^L) \geq \frac{3+\sqrt{5}}{40} \mathcal{D}_{h_0}(G_X^L).} \tag{71}$$

Both statements include the zero-energy case. The selected support is an exact subset of the physical terminal-prime support; no ambient prime has been inserted.

Proof. If $\mathcal{D}_{h_0}(G_X^L) = 0$, take the zero projection. Otherwise apply theorem 3.1 to the finite terminalized vector with the gauges (37). The first conclusion is the diagonal pigeonhole. The second is (47) with $M = 5$, together with (69) and (51). □

The theorem is more restrictive and canonical than selecting one pentant separately in every fiber: remarkably, a single index k_* works in the normalized phase coordinates after the fiber-dependent canonical gauges are removed. In absolute phase coordinates the five windows are rotated by ω_y. The partition itself uses the realized phases $\operatorname{Arg} b_u$.

4.2 Arbitrary fiber families and activation transfer

Let $\mathcal{Y}_* \subseteq \mathcal{Y}_{h_0}^{\mathrm{term}}$ be any declared family of terminal output fibers, and let

$$\mathcal{D}_{h_0}(g; \mathcal{Y}_*) := \sum_{y \in \mathcal{Y}_*} \sum_{u \in \mathcal{F}_y^{\mathrm{term}}} |(D_{h_0} g)_u|^2. \tag{72}$$

Apply the sector theorem after setting every terminalized coordinate outside $\mathcal{Y}_*$ to zero.

Corollary 4.2 (Activation–coverage–sector transfer)**.** *Let g be a literal selected pre-terminal packet with $\mathcal{E}_{\mathrm{pre}}(g) > 0$. Suppose*

$$\mathcal{D}_{h_0}(g) \geq X^{-\lambda_{\mathrm{term}}+o(1)}\mathcal{E}_{\mathrm{pre}}(g), \tag{73}$$

$$\mathcal{D}_{h_0}(g;\mathcal{Y}_*) \geq X^{-\lambda_{\mathrm{fib}}+o(1)}\mathcal{D}_{h_0}(g). \tag{74}$$

Use any exact unit gauges on the active fibers; for the physical vector they may be chosen as in (37)*. Then an adaptive pentant projection supported over $\mathcal{Y}_*$ satisfies*

$$\boxed{\mathcal{G}_{h_0}(P_*g) \geq c_5 X^{-(\lambda_{\mathrm{term}}+\lambda_{\mathrm{fib}})+o(1)}\mathcal{E}_{\mathrm{pre}}(g).} \tag{75}$$

For the family of all active fibers, one has $\lambda_{\mathrm{fib}} = 0$.

Proof. Apply theorem 3.1 to the restriction of $D_{h_0}g$ over $\mathcal{Y}_*$. This gives $\mathcal{G}_{h_0}(P_*g) \geq c_5\mathcal{D}_{h_0}(g;\mathcal{Y}_*)$. Substitute (73) and (74). □

The order of the hypotheses matters. Terminal activation is measured on the full declared pre-terminal packet before any phase selector is formed. The phase selection then acts only on the surviving terminal coordinates. No assertion that a convenient fiber is nonempty or energetically representative is hidden in (74).

4.3 Cofactor blocks and comparison with rank selection

For a cofactor interval $\mathcal{B}_R = [R, 2R]$, put

$$g_R = g\mathbf{1}_{r\in\mathcal{B}_R}. \tag{76}$$

The fibers of $D_{h_0}g_R$ are simply intersections of the full fibers with the block. No new collision is created, so the sector theorem gives

$$\mathcal{G}_{h_0}(P_{*,R}g_R) \geq c_5\mathcal{D}_{h_0}(g_R) \tag{77}$$

for every block, empty or nonempty. Unlike the TPC–55 rank cover, this does not require $R \gg L$ or a bound for the maximal fiber size.

The two selectors solve different problems:

Feature	Prime-rank transversal	Phase-sector selector
Atoms retained per fiber	At most one	Arbitrarily many in one acute sector
Terminal diagonal retained	At least reciprocal maximal multiplicity	At least $1/5$ globally
Grouped/diagonal ratio on selected packet	Exactly one	At least $\cos^2(\pi/5)$
Balanced cofactor input	Needed for subpower loss	Not needed
Dependence on coefficient phases	None	Essential under the current coefficient class
Pre-terminal smooth provenance	Not automatic	Not automatic

Thus (71) improves the retained-energy constant over rank selection at the *adaptive selected* grouping gate, at the price of coefficient adaptivity. It removes the multiplicity cost there, but does not return the deleted four pentants to the full physical output.

4.4 Five-channel form of the remaining cancellation

For the physical vector define

$$z_{y,k} := \sum_{p \in \mathcal{P}^{\mathrm{sec}}_{y,k}} x_{y,p}. \tag{78}$$

Then the original output and the phase-split energy satisfy the exact relations

$$(\mathcal{T}_{h_0} G^L_X)_y = \sum_{k=0}^{4} z_{y,k}, \tag{79}$$

$$\sum_y \sum_{k=0}^{4} |z_{y,k}|^2 \geq \cos^2(\pi/5)\mathcal{D}_{h_0}(G^L_X). \tag{80}$$

Accordingly, complete within-sector prime cancellation is uniformly excluded, with the quantitative lower bound in (80). The undeleted problem is now the explicit five-channel interference in (79). The latter may still vanish by cancellation between different pentants.

5 Adaptive necessity and the pre-terminal phase gate

The constant in theorem 4.1 does not come from a lower singular value of the full grouping map. It comes from using the distinguished vector to choose a favorable coordinate cone. We now show exactly why that dependence is necessary for the universal unrestricted-array guarantee, record the corresponding local fiberwise obstruction in the bounded literal class, and state a sufficient condition for replacing it by a finite fixed pre-terminal library.

5.1 The sharp coefficient-independent alternative

Let Σ be the abstract grouping map (39). For a fixed set $S \subseteq \mathcal{U}$, write P_S for its coordinate projection. Here S is chosen from the fiber geometry and is independent of the input vector.

The following is the unweighted specialization of the TPC–55 coordinate-transversal criterion, recalled here for comparison [6].

Theorem 5.1 (Fixed-selector dichotomy)**.** *The following are equivalent:*

(i) *There exists $c > 0$ such that*

$$\|\Sigma P_S x\|_2^2 \geq c\|P_S x\|_2^2 \qquad (x \in \mathbb{C}^{\mathcal{U}}). \tag{81}$$

(ii) *Every fiber contains at most one selected coordinate:*

$$|S \cap \mathcal{F}_y| \leq 1 \qquad (y \in \mathcal{Y}). \tag{82}$$

When these conditions hold, (81) is an identity with $c = 1$. If they fail, the restricted map has a nonzero two-coordinate kernel vector.

Proof. If (82) holds, every selected output is either zero or one input coordinate, hence $\|\Sigma P_S x\|_2 = \|P_S x\|_2$. Conversely, suppose $u, v \in S \cap \mathcal{F}_y$ are distinct. Take $x_u = 1$, $x_v = -1$, and all other coordinates zero. Then $P_S x = x \neq 0$ but $\Sigma P_S x = 0$, so no positive c is possible. $\square$

Thus a single coefficient-independent selector with a uniform lower bound on the unrestricted coefficient space is exactly a transversal. The pentant theorem escapes this dichotomy because its coordinate set changes with the realized phases. It retains a constant fraction of the energy without claiming a lower bound for all vectors on the same selected set.

5.2 Realization of the local collision in the literal class

The abstract kernel direction is compatible with the source hypothesis actually inherited by the literal packet. Factor the remaining amplitude (35) as

$$b_u = \kappa_u \zeta_{m_u}, \tag{83}$$

where κ_u is the product of every displayed factor other than ζ_{m_u}. Call a terminal atom *baseline nondegenerate* when $\Omega_{q(u)}\kappa_u \neq 0$.

Proposition 5.2 (Literal two-atom saturation)**.** *Suppose $C > 0$ and one exact physical-capable fiber $\mathcal{F}_y^{\mathrm{term}}$ contains two distinct baseline-nondegenerate atoms u_1, u_2. In the inherited coefficient class $|\zeta_m| \leq C$, there are coefficient choices for which the restriction of that fiber to $\{u_1, u_2\}$ has positive terminal diagonal and zero coherent output. More precisely, for a sufficiently small $t \neq 0$, one may take*

$$\zeta_{m_{u_1}} = \frac{t}{\kappa_{u_1}}, \qquad \zeta_{m_{u_2}} = -\frac{t}{\kappa_{u_2}}, \tag{84}$$

and set the source factors belonging to every other row in this fiber to zero. Then

$$D_y = 2|\Omega_y t|^2 > 0, \qquad C_y = 0. \tag{85}$$

Here D_y, C_y refer to the resulting fiber, with all its other rows zeroed.

Proof. Distinct terminal atoms in a fixed fiber have distinct rows. Indeed, the row determines d, and fixed $s = dr$ then determines r; fixed (j, γ) would otherwise give the same input. Thus the two values of ζ_m in (84) are independent coordinates. Choose $|t|$ small enough that both values have modulus at most C. Equations (83) and (36) give terminal amplitudes $\Omega_y t$ and $-\Omega_y t$, proving (85). □

The proposition is a no-go statement for a uniform *fiberwise* lower bound over the inherited bounded coefficient class. It does not assert that the distinguished inherited coefficients realize these choices, that such a two-atom fiber carries global energy, or that other physical output fibers vanish. Those stronger conclusions would require additional support information.

Prime-resolved injectivity, common Möbius phase, a nonzero terminal weight, and even multiplicity two therefore do not imply positive coherence for arbitrary bounded source phases. This is a coefficient space obstruction, not a proof that the intended arithmetic vector cancels.

5.3 A finite fixed phase-provenance criterion

There is a precise way in which the adaptive selector could become compatible with an earlier source decomposition. For an integer $M \geq 2$, let

$$\mathcal{U}^{\mathrm{pre}} = \bigsqcup_{k=0}^{M-1} \mathcal{A}_k \tag{86}$$

be a partition chosen independently of terminal primality and independently of the numerical values of the source coefficients. Its coordinate projections are fixed before D_{h_0} is applied.

Theorem 5.3 (Pre-terminal phase-realization criterion)**.** *Let $g \in \mathbb{C}^{\mathcal{U}^{\mathrm{pre}}}$ and put $x = D_{h_0} g$. Suppose $0 \leq \theta < \pi/2$, and there are angles $\alpha_0, \ldots, \alpha_{M-1}$ and unit output gauges ω_y, all fixed from coefficient-independent pre-terminal and output data before the realized numerical source phases are seen. In particular, neither the angles nor the gauges may be chosen from the realized phases of x. Suppose every nonzero terminal amplitude in $\mathcal{A}_k \cap \mathcal{F}_y^{\mathrm{term}}$ satisfies*

$$\mathrm{dist}_{\mathbb{T}}\left(\mathrm{Arg}(\overline{\omega_y} x_u), \alpha_k\right) \leq \theta. \tag{87}$$

Then one of the fixed *pre-terminal cells, chosen by the least-index energy tie rule after terminal activation, satisfies*

$$\boxed{\|\Sigma D_{h_0} P_{\mathcal{A}_{k_*}} g\|_2^2 \geq \frac{\cos^2\theta}{M} \|D_{h_0} g\|_2^2.} \tag{88}$$

The same statement holds after restriction to a declared output-fiber family, relative to its terminal diagonal energy.

Proof. The fixed cells partition the full pre-terminal coordinates and hence their terminal diagonal energies sum to $\|D_{h_0} g\|_2^2$. One cell retains at least an M^{-1} fraction. On each output fiber its surviving amplitudes obey the acute-sector hypothesis (87). The real-projection proof of (49) gives a factor $\cos^2\theta$. □

This theorem separates two kinds of adaptivity. Choosing one member of a fixed finite library by energy pigeonhole is harmless at the exponent level. Defining the library itself from the realized terminal phases, as in theorem 4.1, is a new post-terminal operation.

The row provenance in TPC–25 permits fixed residue characters and unit-modulus factors inside ζ_m [2]. The interface inherited by TPC–53–55, however, records only $|\zeta_m| \ll 1$. It does not prove that the distinguished ζ_m phases are determined by a fixed finite residue partition. Even if they are, one must also select sign-stable cells for every source and profile factor in (35) and prove that those cells carry the required terminal diagonal energy. Geometric nonempty support alone is insufficient by the concentration obstruction of TPC–54 [5].

Accordingly, theorem 5.3 is the exact next positive criterion, not a property silently attributed to the current packet.

6 Why averaged phases and projective layers do not close the gate

The sector theorem is pointwise in the one physical terminal vector. Two nearby routes produce legitimate positive averages but do not imply a lower bound for the prescribed undeleted output. We record the elementary non-implications because both have appeared upstream in the TPC decomposition.

6.1 Shift averages do not select the prescribed shift

Write $e(t) = e^{2\pi i t}$, and give $\mathbb{T}$ normalized Haar measure. Let $\mathscr{H} \subset \mathbb{Z}$ be a finite set of intercepts containing h_0, and let F_h be vectors in an arbitrary Hilbert space. A completed frozen family may satisfy a lower estimate for $\sum_{h\in\mathscr{H}} \|F_h\|^2$, or equivalently for a Fourier integral. That is not a pointwise estimate at h_0.

Proposition 6.1 (Sharp fixed-member nonlocalization)**.** *Assume $|\mathscr{H}| \geq 2$ and let $w_h \geq 0$ be weights with $w_{h_1} > 0$ for some $h_1 \neq h_0$. There is no positive constant c, depending only on the weights, such that*

$$\|F_{h_0}\|^2 \geq c \frac{\sum_{h\in\mathscr{H}} w_h \|F_h\|^2}{\sum_{h\in\mathscr{H}} w_h} \tag{89}$$

for all vector families. The same failure holds when the right-hand side is represented by Parseval as the L^2-energy of

$$\widehat{F}_w(\alpha) = \sum_{h\in\mathscr{H}} \sqrt{w_h}\, F_h e(h\alpha). \tag{90}$$

Proof. Take $F_{h_0} = 0$, choose $F_{h_1} \neq 0$, and set every other member to zero. The right-hand side of (89) is positive and the left-hand side is zero. Parseval gives

$$\int_{\mathbb{T}} \|\widehat{F}_w(\alpha)\|^2 \, d\alpha = \sum_{h\in\mathscr{H}} w_h \|F_h\|^2, \tag{91}$$

so the Fourier formulation has the same counterexample. □

Consequently, a completed-family theorem transfers to the physical member only if it is already uniform pointwise in h, or if one separately proves a localization estimate such as

$$\|F_{h_0}\|^2 \geq X^{-\lambda_{\mathrm{loc}}+o(1)} \frac{\sum_h w_h \|F_h\|^2}{\sum_h w_h}. \tag{92}$$

TPC–48 proves that its frozen coefficient equals the active coefficient at the one original intercept h_0 [3]. For $h \neq h_0$, the support and coefficients remain frozen from h_0; they are algebraic extensions, not newly reconstructed physical packets. Thus an average over that family is useful only with its frozen-family label until (92) is supplied.

6.2 Layerwise lower squares do not survive signed reassembly

The projective representation also contains a signed or complex reassembly over auxiliary layers. Upper estimates pass through this operation by Minkowski; lower estimates do not.

Proposition 6.2 (Lower-reassembly obstruction)**.** *Let ν be a nonzero finite complex measure on Ω such that $\dim L^2(|\nu|) \geq 2$, and let Z_ω range in a nonzero Hilbert space. There is no positive constant c, depending only on $|\nu|$, such that*

$$\left\| \int_\Omega Z_\omega \, d\nu(\omega) \right\|^2 \geq c \int_\Omega \|Z_\omega\|^2 \, d|\nu|(\omega) \tag{93}$$

for all layer fields.

Proof. Let $\rho = d\nu/d|\nu|$, so $|\rho| = 1$ almost everywhere. Since $\dim L^2(|\nu|) \geq 2$, choose a nonzero $f \in L^2(|\nu|)$ orthogonal to the constants, and choose a nonzero vector v. Put $Z_\omega = \overline{\rho(\omega)} f(\omega) v$. Then

$$\int_\Omega Z_\omega \, d\nu(\omega) = v \int_\Omega f \, d|\nu| = 0, \qquad \int_\Omega \|Z_\omega\|^2 \, d|\nu|(\omega) > 0.$$

□

A favorable projective layer, or even a lower square in every layer, is therefore not a lower square for any signed physical reassembly with $\dim L^2(|\nu|) \geq 2$. A valid transfer would require a separately proved lower frame estimate of the form

$$\left\| \int Z_\omega \, d\nu(\omega) \right\|^2 \geq X^{-\lambda_{\mathrm{reas}}+o(1)} \int \|Z_\omega\|^2 \, d|\nu|(\omega) \tag{94}$$

on the distinguished coefficient field or on a rigorously specified subspace. Such an estimate is not a consequence of bounded total variation.

6.3 The five selected channels still do not lower-bound their sum

The same warning applies internally to (79). The sector theorem proves positive energy before the five channels are reassembled. It does not control the signed cross terms between them. For example, two nonzero channels with values v and $-v$ in the same output coordinate have positive split energy and zero sum. This is compatible with the local literal saturation of proposition 5.2.

There are three honest ways to use the present result downstream:

(i) retain the adaptive pentant as an explicitly selected physical subpacket and prove that every later operation respects that selection;

(ii) verify the fixed pre-terminal phase partition of theorem 5.3; or

(iii) estimate the five-channel cross terms and prove a lower-reassembly inequality for the undeleted distinguished vector.

An average over auxiliary phases can help with the third route only after a pointwise localization estimate identifies the physical phase.

6.4 Parity boundary

No step above supplies a lower-bound sieve for simultaneous primes. The terminal support is taken exactly as inherited; the sector theorem starts after the primality restriction and after the terminal diagonal is formed. It proves that a constant part of an already activated complex vector can be selected into an acute cone. It does not prove that the vector is activated often, that the undeleted phases align, or that a fixed even gap occurs infinitely often. In particular, the result is compatible with the classical parity obstruction [1] and is not reported as a parity-sensitive advance.

7 Endpoint ledger, audit, and the next gate

7.1 The selected coherence exponent

The inherited endpoint scales are

$$Q = X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}, \qquad D = X^{10049/52500+o(1)}, \tag{95}$$

and hence

$$L = Q/D = X^{99979/210000+o(1)}. \tag{96}$$

The constant $c_5 = (3+\sqrt{5})/40$ in (71) is independent of every scale. Therefore the adaptive selected grouping gate has

$$\lambda_{\mathrm{coh}}^{\mathrm{sec}} = 0 \tag{97}$$

relative to terminal diagonal energy. Unlike the coefficient-blind prime-rank certificate in TPC–55, this statement has no factor $1 + L/R$ and remains constant even at $R \asymp J$. Its price is structural adaptivity, not a hidden polynomial exponent.

To place the theorem in the full prospective chain, suppose a parent physical atomic energy is first restricted to a compatible packet and a cofactor block at costs

$$X^{-\lambda_{\mathrm{sel}}+o(1)}, \qquad X^{-\lambda_{\mathrm{blk}}+o(1)}. \tag{98}$$

Suppose terminal activation and any restriction to a declared output-fiber family cost

$$X^{-\lambda_{\mathrm{term}}+o(1)}, \qquad X^{-\lambda_{\mathrm{fib}}+o(1)}. \tag{99}$$

Then corollary 4.2 gives the selected fixed-shift residual-output exponent

$$\lambda_{\mathrm{sel}} + \lambda_{\mathrm{blk}} + \lambda_{\mathrm{term}} + \lambda_{\mathrm{fib}}, \tag{100}$$

with no added positive-power coherence cost.

Formula (100) is valid only for the explicitly adaptive terminal subpacket. It may enter the eventual physical proof chain only if later localization, Gram comparison, boundary truncation, and tail reconstruction preserve that coordinate selection, or if the selector is realized by the fixed pre-terminal criterion of theorem 5.3. This is a logical gate, not an exponent silently set to zero.

For the undeleted physical output, one must instead prove

$$\mathcal{G}_{h_0}(g) \geq X^{-\lambda_{\mathrm{coh}}+o(1)} \mathcal{D}_{h_0}(g). \tag{101}$$

The present paper does not prove (101). If a route uses a completed frozen shift family or signed projective layers, it must also charge λ_{loc} and λ_{reas} from estimates of the forms (92) and (94). The complete prospective undeleted ledger is therefore

$$\boxed{\begin{aligned}\lambda_{\mathrm{sel}} + \lambda_{\mathrm{blk}} + \lambda_{\mathrm{term}} + \lambda_{\mathrm{fib}} + \lambda_{\mathrm{coh}} + \lambda_{\mathrm{loc}} + \lambda_{\mathrm{reas}}&\\ + \lambda_{\mathrm{Gram}} + \lambda_{\mathrm{bdry}} + \lambda_{\mathrm{tail}} \le \frac{1}{400}&.\end{aligned}} \tag{102}$$

For a direct pointwise theorem at the already reassembled physical h_0, no localization or projective-reassembly gate is inserted. At equality in (102), a further error of size $X^{-1/400+o(1)}$ need not be absorbable; strict slack or a closed constant-level comparison remains the safe endpoint condition.

7.2 Six-point physical audit

The proof satisfies the following interface audit.

(A1) *Literal coefficients.* The source weight, bounded complex factor, binary mask, four profiles, cofactor sign, and shaped terminal logarithm are all displayed in (13)– (29). Sector membership is formed from their product, not from a proxy coefficient.

(A2) *Fixed physical shift.* Every terminal equality uses the one specified h_0. Frozen intercepts appear only in the nonlocalization warning of section 6.

(A3) *Physical atomic normalization.* The pre-terminal, terminal diagonal, and grouped energies use raw counting measure in (31)– (33). The terminal amplitude is not absorbed into a modified domain norm.

(A4) *Canonical representation.* The prime coordinate comes from the exact TPC–55 injection. The pentants are the five fixed equal arcs, the common output gauge is (37), and ties use the least index. The support remains explicitly coefficient-adaptive; no projective layer is selected.

(A5) *Actual active support.* The full pre-terminal universe, terminal-admissible universe, and physically nonzero universe are distinct in (18), (22), and (23). Every selected prime belongs to the exact physical prime set $\mathcal{P}_y^{\mathrm{phys}}$.

(A6) *Endpoint budget.* The constant sector step has zero positive polynomial exponent only on the adaptive selected packet. All activation, fiber coverage, localization, reassembly, Gram, boundary, and tail losses remain visible in (102).

7.3 Claim table

Statement	Status
Equal-sector compression and pentant constant	Unconditional finite-dimensional theorem (L0).
Rotating-window and phase-error bounds	Unconditional finite-dimensional refinements (L0).
Literal prime-indexed sector projection	Exact on the fixed-h_0 terminal support under the inherited prime-resolved injection (L1).
Constant-energy selected coherent square	Exact post-terminal adaptive L1 selection; no multiplicity or balanced cofactor cost.
Fixed-selector dichotomy	Exact L0 theorem; fixed uniform lower selectors are precisely transversals.
Literal two-atom saturation	Local coefficient-class L1 no-go, conditional on a nondegenerate two-atom carrier; not a claim about the distinguished coefficients or global output.
Fixed pre-terminal phase realization	Sufficient L0/L1 criterion; its phase and energy hypotheses are not verified for the inherited packet.
Terminal activation and energetic block occupancy	Not proved.
Undeleted fixed-h_0 coherent lower square	Not proved; the five inter-sector channels may still cancel.
Frozen localization, native affine Gram, boundary, and tail	Not proved.
Parity improvement or twin-prime consequence	Not claimed.

In particular, none of the following implications is permitted:

(i) a lower bound for one adaptive pentant does not lower-bound the sum of all five pentants;

(ii) a constant selected fraction of terminal diagonal energy does not prove that terminal activation occurs;

(iii) the common Möbius gauge does not constrain the remaining ζ_m and profile phases;

(iv) a phase-defined terminal set is not automatically a fixed smooth or residue cell available before primality is tested;

(v) an average over frozen shifts or auxiliary Fourier phases does not select the prescribed physical coefficient;

(vi) a favorable projective layer does not survive signed reassembly by a lower Minkowski inequality; and

(vii) the residual grouped norm is not automatically the native affine physical Gram.

7.4 The next gate

Complete within-sector cancellation is now uniformly excluded on a constant-energy subpacket. The next stage should investigate the *undeleted* five-channel problem by routes that are still pointwise in h_0. Three concrete quantities are available.

Door A. Measure the terminal diagonal carried by singleton fibers. Their coherent square equals their diagonal without any phase hypothesis. Equivalently, bound the energy-weighted collision incidence of multiple fibers.

Door B. Use the distinct terminal primes to form an auxiliary prime-modulated polynomial, compute its exact Parseval diagonal, and quantify the microscopic localization needed to return its average energy to the physical phase zero.

Door C. Audit whether the actual ζ_m and sign-stable source profiles admit the fixed finite phase partition in theorem 5.3; if not, prove a saturation theorem for every such fixed partition.

Door A is phase-blind but needs arithmetic occupancy. Door B has an exact mean square but a pointwise localization barrier. Door C would convert the present adaptive theorem into a transportable pre-terminal selection. A negative theorem for any door remains useful and must retain its own label.

Conclusion. The literal coherent-prime gate admits a universal constant-energy escape from its collision kernel: five gauge-normalized relative phase sectors suffice, and one global normalized-sector index retains at least $(3+\sqrt{5})/40$ of the full terminal diagonal after grouping. The absolute windows may rotate with the fibers. This removes every positive polynomial multiplicity cost from the adaptive selected square. The same theorem also exposes its exact limitation. On the unrestricted coordinate space, uniform coefficient-independent lower selection is possible only on collision-free transversals. The literal two-atom result proves the corresponding local fiberwise obstruction, not a global coefficient-class no-go. The undeleted five-channel interference, terminal activation, fixed-shift localization, native-Gram comparison, boundary, and tail gates remain open, strictly before any parity-sensitive or twin-prime conclusion.

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